BALL BALANCED ON A BEAM USING AN ESP32 MICROCONTROLLER

Requirement presentation

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Introduction

Balancing a ball on a beam using an ESP32 microcontroller is a classic control systems project that demonstrates real-time feedback and actuation

An ESP32 microcontroller receives data from a distance or position sensor that tracks the ball’s location along the beam. This data feeds into a control algorithm, typically a Proportional-Integral-Derivative (PID) controller, that calculates necessary adjustments to the beam’s tilt. An actuator, such as a servo motor, then tilts the beam accordingly, counter-acting the ball’s movement and maintaining its position.

Objectives

The goal is to keep a ball steady at a specific point on a horizontal beam by adjusting the beam's angle in response to the ball's position.

Requirements

- Functional requirements:

1. Ball Position detection: The position sensor detects the ball's location on the beam, providing input data for the control algorithm

➢ **Ultrasonic** or **Infrared distance sensor**, or a **camera system**, to detect the ball's position accurately

2. Control Layer :

➢ **ESP32 Microcontroller**: The ESP32 serves as the core processing unit

✓ It receives the position data from the sensor

✓ Runs control Algorithms to determine the ball’s current and target position

- ✓ Calculates the required beam angle adjustment to minimize the position error
- **PID Control Algorithms**: Calculates the proportional, integral and derivative responses based on the position error, generating a control output for the actuator to balance the ball effectively.

3. Actuation units:

- **Servo Motor**: The servo motor adjusts the beam's tilt angle based on the control signal output from the ESP32. A precise and fast-acting motor is essential for responding to changes in the ball's position in real-time
- **Motor Driver (optional)**: If the servo motor's power requirements exceed the ESP32's capabilities, a motor driver might be added to manage the motor's current and voltage.

4. Power supply:

- **Power source:** Provides stable power to the ESP32, sensors, and motor, typically a regulated DC power supply or battery.
- **Voltage Regulation**: Ensures that each component receives the appropriate voltage level for reliable operation.

5. Feedback and Communication Layer:

- **Feedback loop:** The system operates in a closed-loop, where the ball's position data continuously updates the control loop, allowing real-time adjustments to the beam's angle
- **User Interface (optional)**: A dashboard or allows users to monitor the ball's position, adjust PID parameters, or view real-time performance data.

6. Software requirement:

- ✓ C++
- ✓ Simulink
- ✓ Arduino code

- Non-functional requirements

1. **Performance**: The system should respond to changes in the ball's position in under 100 milliseconds, maintaining a high level of responsiveness to ensure real-time balancing.
2. **Stability**: The system should handle minor disturbances (e.g., small vibrations) without losing stability.
3. **Scalability (Adaptability)**: The control algorithm should be adaptable for different beam lengths, ball sizes, or sensor types with minimal code or hardware adjustments.
4. **Environmental Tolerance**: It should be able to tolerate small environmental changes without performance degradation

Limitaions

1. Sensor Limitations
2. Actuator (Servo Motor) Limitations
3. Control Algorithm Limitations : PID Tuning Complexity
4. Power Supply Constraints: Battery Life, Voltage and Current Stability
5. Physical and Mechanical Limitations
6. Environmental Influences
7. Real-Time Feedback and Communication Delays

Conclusion

This project is ideal for exploring feedback loops, sensor integration, and precise motor control. The ESP32’s processing speed and real-time capabilities make it well-suited for handling the continuous adjustments required to achieve stable balance

THANKS